

Theoretical Astrophysics: Stellar Structure and Evolution

Homework 3

P. Käpylä, I. Milić

July 15, 2025

Deadline for this homework is 30/07 23:59

Problem 1: Building on the hands-on 6&7, conduct the following numerical calculations/experiments, to investigate 3D simulation of convection in more detail (*Note:* some of these calculations were already covered in the lecture).

Important: Please do not turn in jupyter notebook, but rather a textual report, in pdf format or similar.

- Try to define the surface of the simulated atmosphere. Discuss the result.
- Assuming the atmosphere is composed of pure hydrogen (it is not, but for simplicity assume this), solve the equation of state for one fixed x and y , and different z . That is, using Saha equation and results of hands-on 5, estimate the mass density from the values of temperature and pressure. Note that the pressure provided in the simulation is in CGS units!
- Repeat this calculation for the whole atmosphere. *Alternatively: choose a reasonable number of mean molecular mass to find mass density from the pressure and temperature without solving for equation of state explicitly.*
- Calculate the kinetic energy flux density, and mean kinetic energy flux density (averaged over x and y). Plot mean kinetic energy flux density as a function of z . Recall that kinetic energy flux density is:

$$F_{kin} = \frac{\rho v^2}{2} v_z \quad (1)$$

- From the fluctuations of temperature and momentum density, calculate the enthalpy flux density and mean enthalpy flux density

$$F_{ent} = (\rho v_z - \langle \rho v_z \rangle_{x,y}) C_P (T - \langle T \rangle_{x,y}) \quad (2)$$

Where for C_P you can take the value 21 kJ / kg/ K.

-
- Analyze the behavior of mean enthalpy flux density and mean kinetic flux density, as well as their sum , with height. What can you conclude?

Useful physical constants

- $R_{\odot} = 696 \times 10^6 \text{ m}$
 - $M_{\odot} = 1.989 \times 10^{30} \text{ kg}$
 - $L_{\odot} = 3.83 \times 10^{26} \text{ W}$
 - $T_{\odot}^{\text{eff}} = 5777 \text{ K}$
 - $1 \text{ AU} = 1.496 \times 10^8 \text{ km}$
 - $c = 2.997 \times 10^8 \text{ m/s}$
 - $G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
 - $k = 1.38 \cdot 10^{-23} \text{ J/K}$
 - $m_{\text{H}} = 1.67 \cdot 10^{-27} \text{ kg}$
 - $h = 6.626 \times 10^{-34} \text{ J s.}$
 - $k = 1.38 \times 10^{-23} \text{ J/K.}$
-